

The Bishop-Phelps-Bollobas Moduli of $M_2(\mathbb{C})$ Are Maximal

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Source Question

The source paper is Chica–Kadets–Martin–Moreno–Pulido–Rambla–Barreno, *Bishop-Phelps-Bollobas moduli of a Banach space*, arXiv:1304.0376. It proves that many Banach spaces have maximal Bishop-Phelps-Bollobas moduli, including unital C^* -algebras with nontrivial center, and then raises the case of $L(H)$, where those centralizer methods do not apply.

BISHOP-PHELPS-BOLLOBÁS MODULI OF A BANACH SPACE

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It would be interesting to see whether the algebra $L(H)$ for a finite- or infinite-dimensional Hilbert space H has the maximum Bishop-Phelps-Bollobás moduli. None of the results of this section applies to it since its center is trivial and, despite it contains $K(H)$ as an M -ideal, there is no element $x_0 \in L(H)$ satisfying the requirements of Corollary 4.9 (see [?, page 538]). Let us also comment that the bidual of $L(H)$ is a C^* -algebra with non-trivial centralizer, so $\Phi_{L(H)^{**}}(\delta) = \Phi_{L(H)^{**}}^S(\delta) = \sqrt{2}\delta$ for every $\delta \in (0, 1/2]$. If there is $\delta \in (0, 1/2]$ such that $\Phi_{L(H)}(\delta) < \sqrt{2}\delta$, then this would be an example when the inequality in Proposition 3.6 is strict.

We finish this section with two pictures: one with the Bishop-Phelps-Bollobás moduli of $\mathbb{R}$, $\mathbb{C}$ and $\ell_\infty^{(2)}$, and another one with the corresponding values of the spherical Bishop-Phelps-Bollobás moduli.



This packet proves the first noncentral Hilbert-operator case:

$$L(\mathbb{C}^2) = M_2(\mathbb{C}).$$

It does not solve the higher-dimensional or infinite-dimensional $L(H)$ problem.

Definitions

For a Banach space X , write

$$\Pi(X) = \{(y, y^*) \in S_X \times S_{X^*} : y^*(y) = 1\}.$$

The Bishop-Phelps-Bollobas modulus $\Phi_X(\delta)$ is the least radius which approximates every pair $(x, x^*) \in B_X \times B_{X^*}$ satisfying $\operatorname{Re} x^*(x) > 1 - \delta$ by some element of $\Pi(X)$. The spherical modulus $\Phi_X^S(\delta)$ is defined in the same way but only for $(x, x^*) \in S_X \times S_{X^*}$.

The source paper proves the universal bound

$$\Phi_X^S(\delta) \leq \Phi_X(\delta) \leq \sqrt{2}\delta \quad (0 < \delta < 2)$$

and also proves that $\Phi_X(\delta) = \sqrt{2\delta}$ if and only if $\Phi_X^S(\delta) = \sqrt{2\delta}$. We only need the upper bound and the obvious implication from a spherical lower bound to an ordinary lower bound.

We identify $M_2(\mathbb{C})^*$ with S_1^2 , the trace-norm class, by

$$T(a) = \text{Tr}(Ta), \quad \|T\|_{(M_2)^*} = \|T\|_1.$$

Theorem

Theorem 1. *Let $X = M_2(\mathbb{C})$ with the operator norm. Then*

$$\Phi_X(\delta) = \Phi_X^S(\delta) = \sqrt{2\delta} \quad (0 < \delta \leq 1/2).$$

Proof

We first record the trace-norm obstruction.

Lemma 1. *Let $0 < \varepsilon \leq 1$, put*

$$x = \text{diag}(1 - \varepsilon, 1), \quad \rho = \text{diag}(\varepsilon/2, 1 - \varepsilon/2).$$

If $(y, T) \in \Pi(M_2(\mathbb{C}))$, then

$$\max\{\|x - y\|, \|\rho - T\|_1\} \geq \varepsilon.$$

Proof. Let $(y, T) \in \Pi(M_2(\mathbb{C}))$, so

$$\|y\| = 1, \quad \|T\|_1 = 1, \quad \text{Tr}(Ty) = 1.$$

If $\|x - y\| \geq \varepsilon$, there is nothing to prove. Assume from now on that $\|x - y\| < \varepsilon$.

First, y is not unitary. Indeed, for every unitary u ,

$$\|x - u\| \geq \|xe_1 - ue_1\| \geq \|ue_1\| - \|xe_1\| = 1 - (1 - \varepsilon) = \varepsilon.$$

Thus $\|x - y\| < \varepsilon$ excludes unitary y .

Since y is a nonunitary 2×2 contraction of norm one, its singular values are 1 and s , where $0 \leq s < 1$. Choose a singular value decomposition

$$y = U \text{diag}(1, s) V^*.$$

Set $S = V^* T U$. Then $\|S\|_1 = 1$ and, by cyclicity of the trace,

$$1 = \text{Tr}(Ty) = \text{Tr}(S \text{diag}(1, s)).$$

Writing $S = (s_{ij})$, we get

$$1 = |s_{11} + s s_{22}| \leq |s_{11}| + s |s_{22}| \leq |s_{11}| + |s_{22}| \leq \|S\|_1 = 1.$$

Because $s < 1$, equality forces $s_{22} = 0$ and $|s_{11}| = 1$. The trace norm dominates the Hilbert-Schmidt norm, so

$$1 = \|S\|_1 \geq \|S\|_2 = (|s_{11}|^2 + |s_{12}|^2 + |s_{21}|^2 + |s_{22}|^2)^{1/2}.$$

Since $|s_{11}| = 1$, this gives $s_{12} = s_{21} = 0$. Hence $S = \text{diag}(1, 0)$, after the phase is fixed by $\text{Tr}(S \text{diag}(1, s)) = 1$, and therefore T is rank one.

Now compare ρ with rank-one trace-norm-one matrices. The singular values of ρ are $1 - \varepsilon/2$ and $\varepsilon/2$, while the singular values of any rank-one T with $\|T\|_1 = 1$ are 1 and 0. The Ky Fan–Mirsky singular-value inequality gives

$$\|\rho - T\|_1 \geq |(1 - \varepsilon/2) - 1| + |\varepsilon/2 - 0| = \varepsilon.$$

Therefore, if $\|x - y\| < \varepsilon$, then $\|\rho - T\|_1 \geq \varepsilon$. This proves the lemma. $\square$

We now prove the theorem. Fix $0 < \delta \leq 1/2$. Let $0 < \gamma < \delta$ and set $\varepsilon = \sqrt{2\gamma}$. Since $\gamma < \delta \leq 1/2$, we have $0 < \varepsilon \leq 1$. With x and ρ as in Lemma 1,

$$\|x\| = 1, \quad \|\rho\|_1 = 1,$$

and

$$\text{Tr}(\rho x) = (\varepsilon/2)(1 - \varepsilon) + (1 - \varepsilon/2) = 1 - \varepsilon^2/2 = 1 - \gamma > 1 - \delta.$$

Thus (x, ρ) is an admissible spherical test pair for $\Phi_X^S(\delta)$. Lemma 1 says that its d_∞ -distance to $\Pi(X)$ is at least $\varepsilon = \sqrt{2\gamma}$. Hence

$$\Phi_X^S(\delta) \geq \sqrt{2\gamma}.$$

Letting $\gamma \uparrow \delta$, we obtain

$$\Phi_X^S(\delta) \geq \sqrt{2\delta}.$$

The universal upper bound gives

$$\Phi_X^S(\delta) \leq \Phi_X(\delta) \leq \sqrt{2\delta}.$$

Therefore both moduli are equal to $\sqrt{2\delta}$.

Limitations and Next Target

This proof is genuinely two-dimensional. The step “ y is nonunitary, hence the top singular subspace is one-dimensional” fails in M_n for $n \geq 3$: a nonunitary norm-one contraction may still have a large singular subspace for singular value 1. The higher-dimensional noncentral-projection route would need to control norming trace-class functionals supported on that larger top singular face.

Thus this packet gives a verified base case for the source paper’s $L(H)$ question, not the full answer for finite-dimensional H nor for infinite-dimensional $B(H)$.

References

- [1] M. Chica, V. Kadets, M. Martin, S. Moreno-Pulido and F. Rambla-Barreno, *Bishop-Phelps-Bollobas moduli of a Banach space*, J. Math. Anal. Appl. 412 (2014), 697–719; arXiv:1304.0376.
- [2] R. Bhatia, *Matrix Analysis*, Graduate Texts in Mathematics 169, Springer, 1997.